Product Requirements Document

AI-Powered Vinyl Wrapping Visualization System

A commercial SaaS platform for realistic, AI-driven vinyl wrap previews

Field	Detail
Document Owner	Product Owner (TBD)
Status	Draft v1.0
Date	August 1, 2026
Target Release	Phase 1 – MVP

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1. Document Overview

This Product Requirements Document (PRD) defines the purpose, scope, features, and success criteria for the AI-Powered Vinyl Wrapping Visualization System — a commercial SaaS platform sold to vinyl wrapping companies. It translates the project definition into actionable product requirements for design, engineering, and go-to-market teams.

2. Executive Summary

The AI-Powered Vinyl Wrapping Visualization System lets vinyl wrapping businesses show customers a photorealistic preview of how a chosen vinyl design will look on a real surface — a wall, door, cabinet, vehicle panel, or similar — before any physical installation happens. Customers upload a photo of the target surface, select the surface region, choose a vinyl from the company's catalogue, and the system generates a preview that replaces only that surface while preserving the room's geometry, lighting, shadows, and surrounding objects.

The product's core value proposition is confidence: customers commit to a design because they can see it first, which shortens sales cycles, cuts down on returns and rework, and reduces the manual editing effort currently spent producing mockups by hand.

3. Problem Statement

Vinyl wrapping companies currently rely on printed catalogues, small physical swatches, or manually edited photos to help customers visualize a finished wrap. This creates friction across the sales funnel:

- Customers struggle to imagine the final result, leading to hesitation and delayed decisions.
- Manual mockups are slow and expensive, and don't scale across a large catalogue of vinyl designs.
- Mismatched expectations between what customers approve and what installers deliver lead to rework and returns.
- Sales representatives lack a fast, repeatable way to present multiple design options during a single consultation.

The system directly addresses this by generating realistic, catalogue-accurate previews in the time it takes to browse a product list.

4. Goals and Objectives

4.1 Business Goals

- Reduce the average sales cycle by giving customers a fast, confident basis for decision-making.
- Reduce manual visualization/design labor per customer inquiry.
- Reduce post-installation returns and design changes caused by mismatched expectations.
- Establish a licensable, multi-tenant platform that can be sold to vinyl wrapping companies as a subscription product.

4.2 Success Metrics

Metric	Target (Phase 1)
Perceived visual realism (user rating)	> 90%
Time to generate a visualization	Shortest practical time; target under system-defined SLA
Surfaces correctly segmented without manual correction	High accuracy across common surface types (wall, door, cabinet, floor)
Reduction in manual mockup effort vs. current process	Significant reduction, measured post-launch
Customer-reported confidence in design choice	Improved vs. baseline (catalogue-only) process

5. Target Users & Personas

Persona	Description	Primary Needs
Homeowner / Customer	Uploads a photo of their space or item, browses the vinyl catalogue, and previews designs to decide on a purchase.	Fast, realistic preview; easy comparison across designs; simple upload/selection flow.
Sales Representative	Uses the tool during in-person or remote consultations to generate options and prepare quotations.	Quick multi-design generation; ability to save and share visualizations; minimal training overhead.
Vinyl Installer	References the approved visualization to understand scope and confirm expectations before starting work.	Clear, accurate representation of the approved surfaces and material; confirmation of customer sign-off.

6. Scope

6.1 In Scope (Phase 1)

- Upload of a room/surface photograph at any resolution.
- Manual selection of one or more predefined room components (wall, floor, door, cabinet, wardrobe, ceiling, furniture, kitchen counter, bathroom wall, glass panel, window frame, reception desk).
- Selection of a vinyl design from the company's digital catalogue.
- AI-driven surface segmentation, scene understanding, texture mapping, rendering, and post-processing.
- Generation of a single photorealistic visualization preserving room geometry, lighting, shadows, and surrounding objects.
- Saving, downloading, comparing, and re-editing previous visualizations.

6.2 Out of Scope (Phase 1)

- Interior redesign or furniture rearrangement.
- Text-to-image generation of new scenes.
- Automatic room decoration.
- Creation of new vinyl textures not present in the catalogue.
- Augmented reality, video-based visualization, and 3D room reconstruction (deferred to future phases).

7. User Stories & Use Cases

ID	As a...	I want to...	So that...
US-01	Homeowner	upload a photo of my wall/door/cabinet	I can preview vinyl designs on my actual space.
US-02	Homeowner	select multiple surfaces in one photo	I can see a coordinated look across several components at once.
US-03	Homeowner	browse and apply different catalogue vinyls to the same photo	I can compare options before deciding.
US-04	Homeowner	download or save the final visualization	I can share it with family or revisit it later.
US-05	Sales Representative	generate several visualizations quickly during a consultation	I can help the customer decide in a single meeting.
US-06	Sales Representative	prepare a quotation alongside the visualization	I can move the customer toward a purchase decision.
US-07	Vinyl Installer	view the approved visualization before starting work	I understand exactly which surfaces and materials were approved.
US-08	Homeowner	choose whether fixtures like windows or sockets are included in the wrap	the preview matches what I actually want installed.

8. Functional Requirements

ID	Requirement	Priority
FR-01	Users can upload a room/surface image in common formats and resolutions; the system normalizes size, orientation, and color automatically.	Must
FR-02	Users can select one or more predefined room components for a given uploaded image.	Must
FR-03	Users can browse a digital vinyl catalogue with Vinyl ID, product name, high-resolution texture, and color code per entry.	Must
FR-04	The system segments the selected component(s) with a precise mask before material replacement.	Must
FR-05	The system estimates depth, perspective, and surface orientation to correctly project the vinyl texture.	Must
FR-06	The system blends the projected material into the original image while preserving lighting, shadows, reflections, and edges.	Must
FR-07	The system refines mask boundaries and removes artifacts in a post-processing step before final output.	Must
FR-08	Users can preview and compare multiple vinyl designs against the same uploaded image.	Must
FR-09	Users can save a visualization project and return to re-edit it later.	Should

ID	Requirement	Priority
FR-10	Users can download the final high-resolution visualization.	Must
FR-11	Users can include or exclude fixtures (windows, sockets, mirrors) from the wrapped area.	Should
FR-12	Users can undo/redo selection and design changes within a session.	Should
FR-13	The system supports applying vinyl to multiple selected components within a single generation request.	Should
FR-14	Texture rotation control for applied vinyl.	Could (future enhancement)

9. Non-Functional Requirements

Category	Requirement
Realism	Maintain greater than 90% perceived visual realism for the initial release.
Fidelity	Preserve catalogue colors and textures accurately; only the selected material should visibly change.
Performance	Produce visualization results within the shortest practical time for a good consultation-time experience.
Scalability	Support cloud-native deployment with independent scaling of image processing, AI inference, rendering, storage, and user management.
Modularity	AI models must be replaceable/upgradable independently without affecting the rest of the platform.
Compatibility	Handle different room and surface types and support multiple selected surfaces per request.
Extensibility	Architecture must accommodate future features (AR, 3D reconstruction, video) without a major redesign.

10. Expected Workflow

1. User uploads a room or surface photograph.
2. User selects one or more room components.
3. User selects one or more vinyl designs from the catalogue.
4. The AI identifies and extracts the selected components (segmentation).
5. The AI analyzes room geometry and camera perspective (scene understanding).
6. The selected vinyl texture is projected onto the extracted surfaces (texture mapping).
7. The rendering engine blends the material while preserving lighting, perspective, shadows, and object boundaries.
8. Optional fixtures (windows, sockets, etc.) are included or excluded per user preference.
9. The system produces the final high-quality visualization.
10. The user compares different vinyl designs before making a purchase decision.

11. AI Workflow & System Architecture

The platform follows a modular microservices architecture, where each AI capability is an independently deployable, independently scalable service communicating through well-defined APIs.

Module	Responsibility
Image Processing	Normalize uploaded image size, orientation, and color before downstream processing.
Surface Segmentation	Identify the selected room component(s) and generate a precise segmentation mask.
Scene Understanding	Estimate depth, perspective, and surface orientation.
Texture Mapping	Project the selected vinyl texture onto the segmented surface, preserving geometry and scale.
Rendering	Blend the projected material with the original image, preserving lighting, shadows, reflections, and edges.
Post-Processing	Refine boundaries, remove artifacts, improve sharpness, and produce the final visualization.

AI Philosophy: the system performs exactly one operation — replacing the selected surface material with the selected vinyl while preserving the original scene. It must never redesign the room, generate a wholly new image, or alter furniture placement, wall shape, room dimensions, camera angle, or environmental lighting.

12. Inputs & Data Model

12.1 Room Image

A photograph captured via mobile phone or camera, at any resolution. The system must automatically process varying resolutions.

12.2 Selected Components

Users manually choose one or more components from a predefined list, including: wall, floor, door, cabinet, wardrobe, ceiling, furniture, kitchen counter, bathroom wall, glass panel, window frame, and reception desk.

12.3 Vinyl Catalogue Entry

Field	Description
Vinyl ID	Unique identifier for the catalogue entry.
Product Name	Display name of the vinyl design.
Texture Image	High-resolution source texture used for mapping.
Color Code	Reference color code for the material.
Finish Type	Future enhancement — matte, gloss, textured, etc.
Additional Metadata	Future enhancement — e.g., pricing, availability.

13. User Controls

- Manually select surfaces to be wrapped.
- Apply different vinyl styles to the same selection.
- Rotate applied textures (future enhancement).
- Replace multiple surfaces within one request.
- Choose whether windows, sockets, mirrors, or other fixtures are included in the wrap.
- Undo and redo changes within a session.
- Compare before/after images side by side.

14. Technical Design Principles

- Modular microservices architecture with single-responsibility services.
- AI models are replaceable without affecting the rest of the platform.
- Inter-service communication through well-defined APIs.
- Independent scaling of image processing, AI inference, rendering, storage, and user management.
- Cloud-native deployment with GPU acceleration for AI inference.
- Architecture designed for future expansion without major rework.

15. Assumptions & Constraints

- Vinyl catalogue content (texture images, color codes) is supplied and maintained by each tenant company.
- Users have access to a camera-equipped device capable of capturing a usable room/surface photo.
- Initial release targets common indoor and vehicle-adjacent surface types; highly irregular or reflective surfaces may have reduced accuracy.
- Phase 1 excludes AR, video, and 3D reconstruction; these are explicitly deferred to future phases.

16. Risks & Mitigations

Risk	Mitigation
Segmentation inaccuracy on complex or cluttered scenes	Provide manual mask correction tools; continuously expand training data for edge cases.
Unrealistic texture projection on curved or angled surfaces	Invest in robust depth/perspective estimation; support manual perspective adjustment as a fallback.
Slow generation time impacting in-consultation usability	Optimize inference pipeline; use GPU-accelerated cloud infrastructure; cache common operations.
Catalogue color/texture fidelity loss during rendering	Apply color-accurate rendering pipelines and validate against source texture references.
Scaling costs as tenant and image volume grow	Modular, independently scalable services; usage-based cloud resource allocation.

17. Future Roadmap

- Automatic room measurement and vinyl quantity estimation.
- Cost estimation based on measured area and selected material.
- AI-based vinyl design recommendations.
- Augmented reality visualization.
- Video-based visualization.
- 3D room reconstruction.
- Enterprise dashboards and reporting.
- ERP and CRM integrations.
- Installer planning tools.
- Inventory-aware catalogue management.

18. Success Criteria

- Produces realistic visualizations suitable for customer approval.
- Preserves the visual characteristics of the selected vinyl catalogue material.
- Modifies only the user-selected surfaces.
- Handles multiple room types and multiple selected components.
- Reduces manual visualization effort compared to the current process.
- Improves customer confidence during the sales process.
- Provides a scalable foundation for future AI and enterprise features.

19. Open Questions

- What is the target time-to-first-visualization SLA for Phase 1 (e.g., under 30 seconds, under 2 minutes)?
- Will Phase 1 support vehicle surfaces (e.g., car panels) in addition to interior/architectural surfaces, or is that a later phase?
- Who owns catalogue data ingestion and quality control — the platform team or each tenant company?
- What pricing/licensing model will be used across tenant companies (per-seat, per-visualization, flat subscription)?